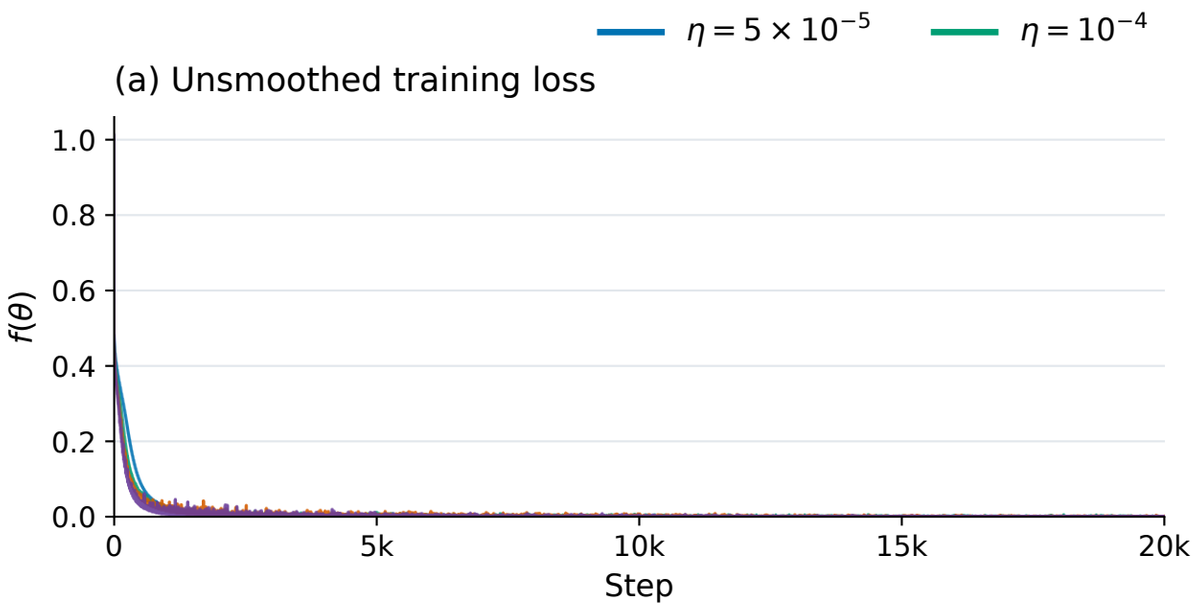
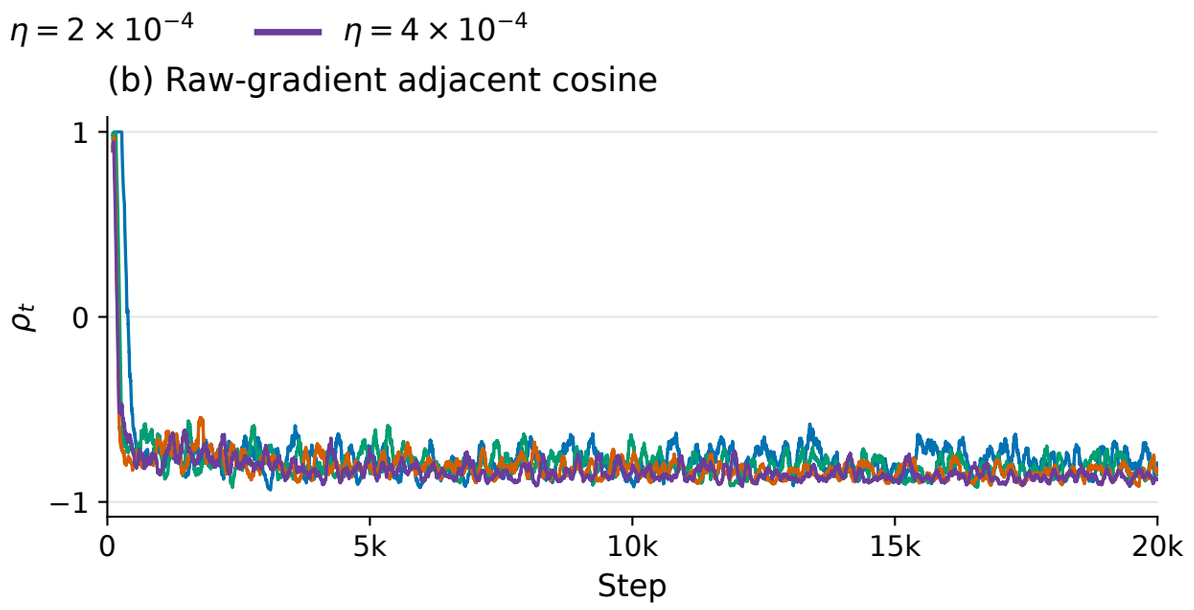
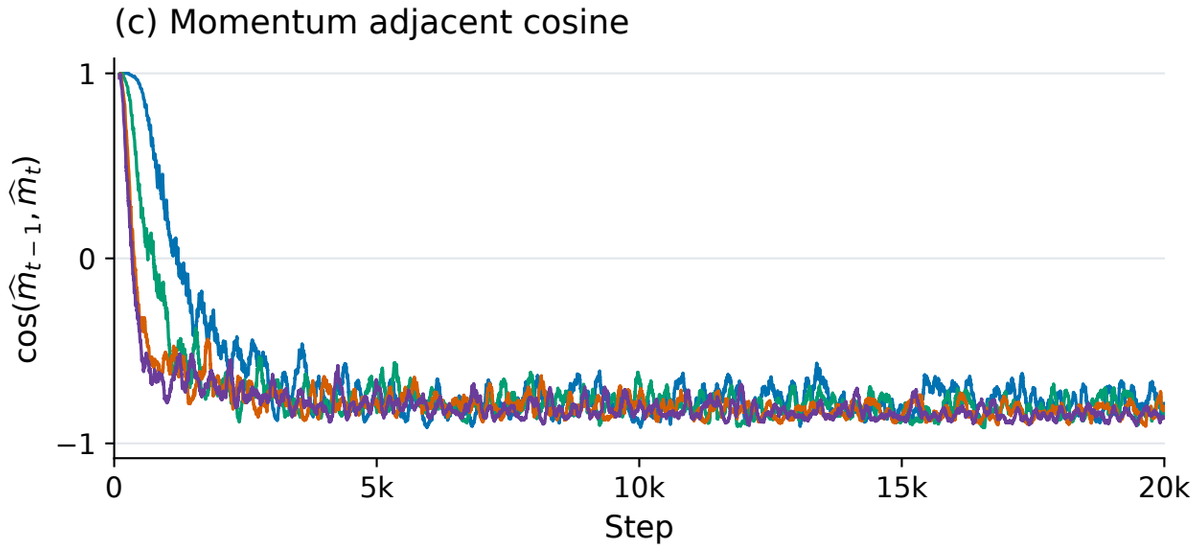
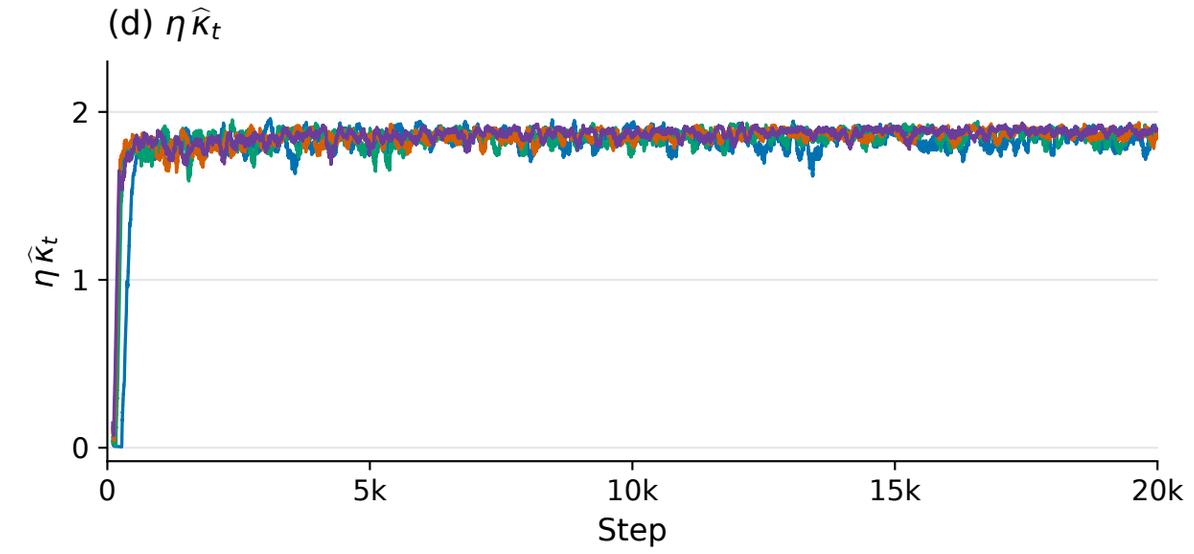
# CIFAR-10 | Full-batch Adam

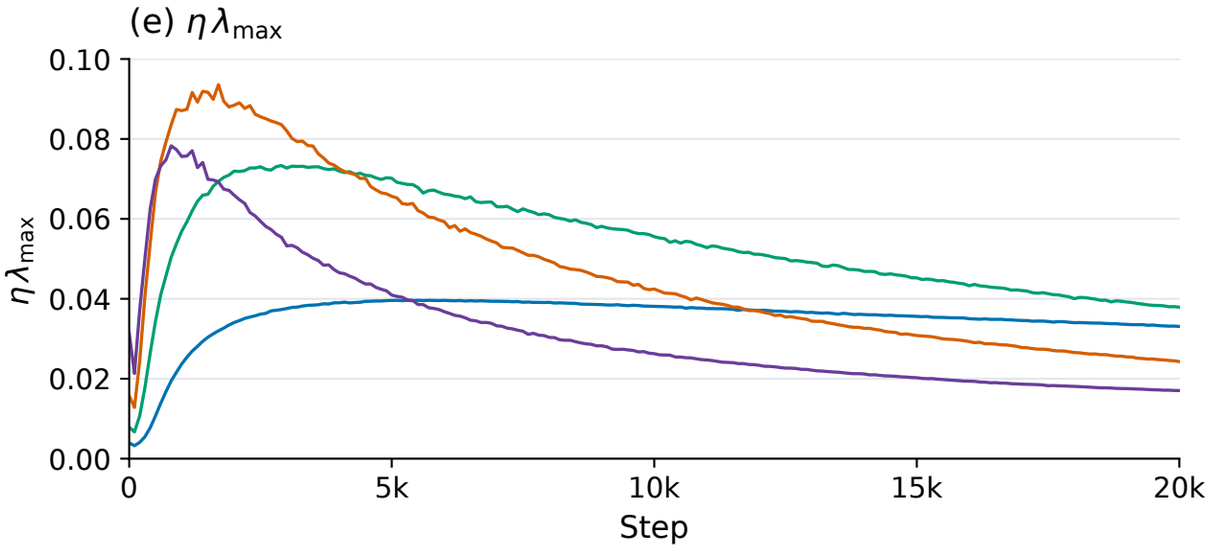
5,000 training images | tanh 200-200 MLP | 20,000 updates

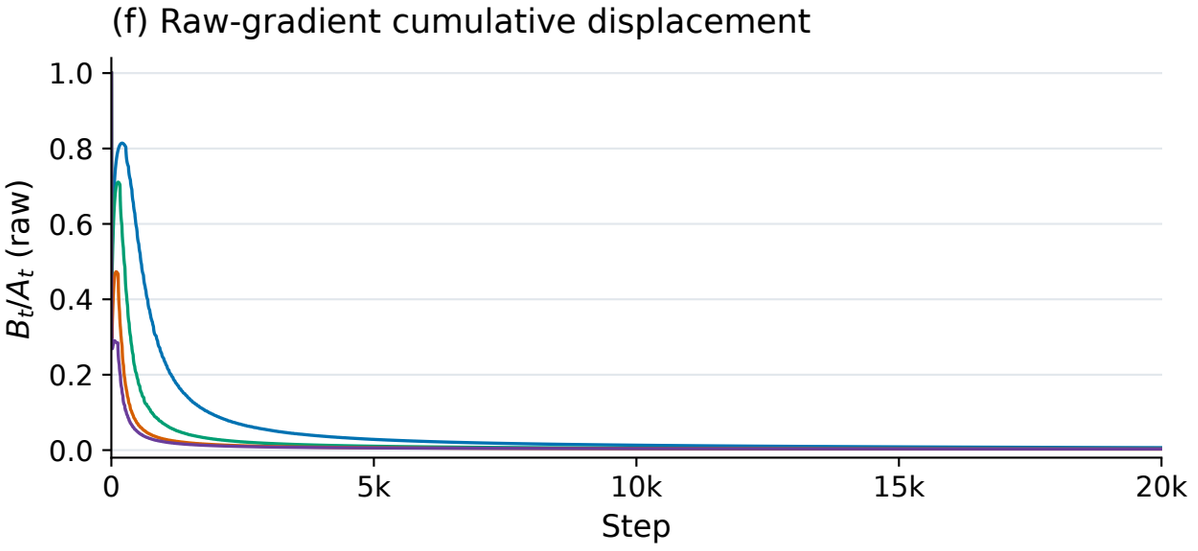


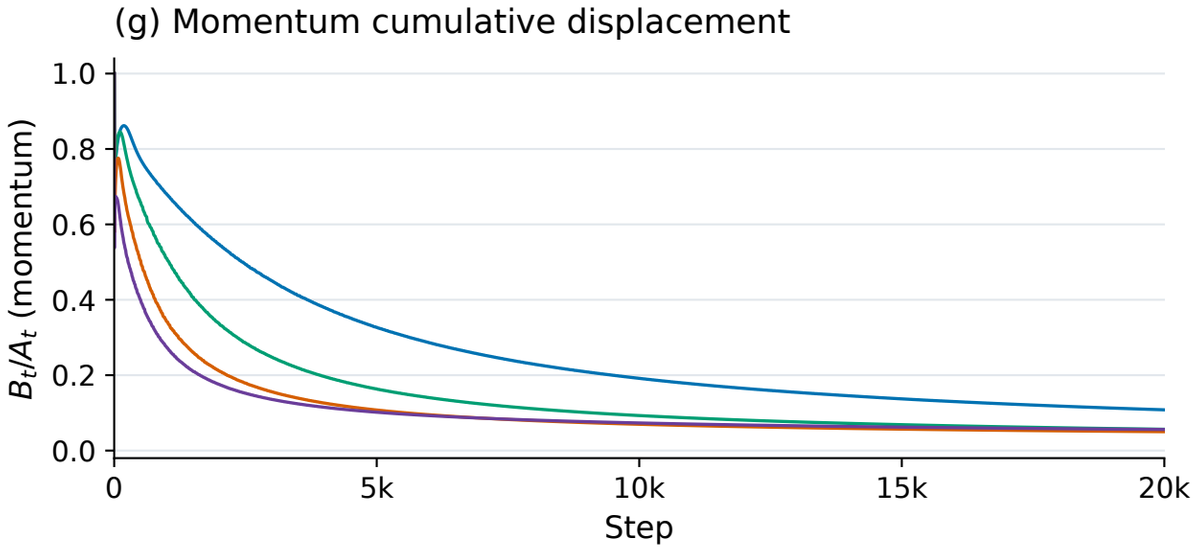



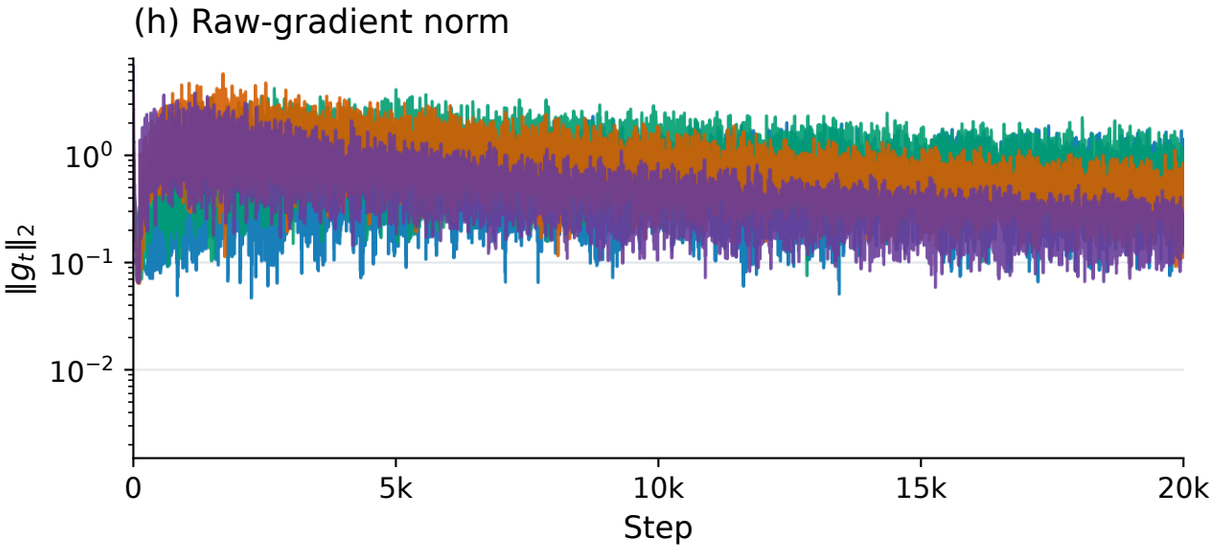



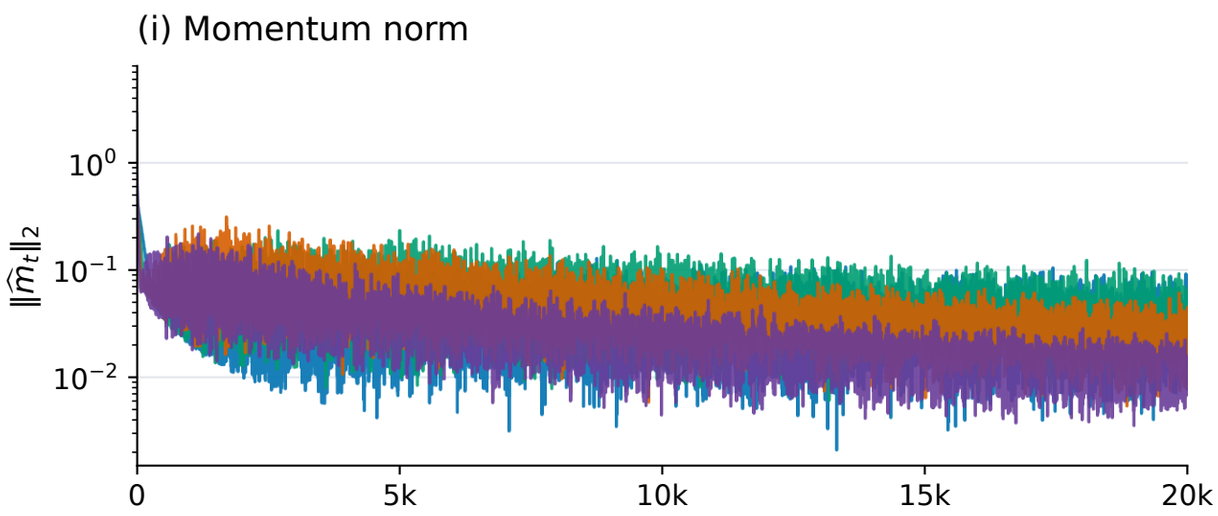



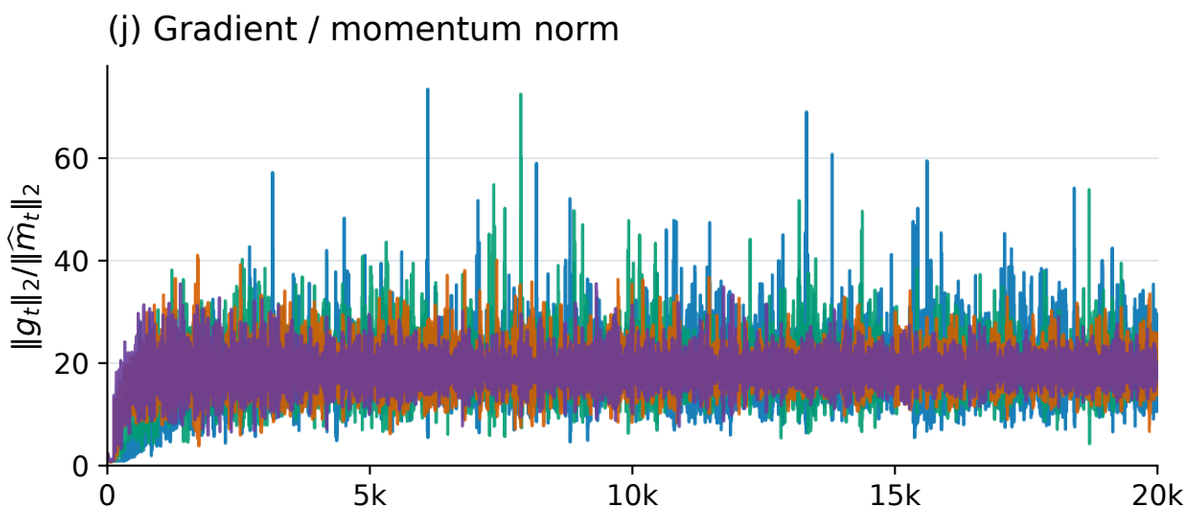




Momentum is bias-corrected: $\widehat{m}_t = m_t/(1-0.9^t)$. Cosines and $\eta\widehat{\kappa}$: 100-pair trailing means.

Loss, norms and norm ratio are unsmoothed. Only norm panels use log Y; full and early views share Y scales.

Ordinary Hessian every 100 steps. $\eta\widehat{\kappa}$ is a gradient trajectory proxy along Adam.